# Cutaneous and Systemic Manifestations of Hypothyroidism

The skin in hypothyroidism is typically **xerotic**, with a poorly hydrated **stratum corneum**. Histologically, the epidermis is thin, showing **hyperkeratosis** and **follicular plugging**. These changes are generalized in distribution, which helps differentiate them from conditions such as **atopic dermatitis** and **keratosis pilaris**, which predominantly affect the extremities. Additionally, a **fine wrinkling** of the skin may be observed.

A distinctive **yellow-orange discoloration** of the skin—known as **carotenemia**—can occur due to accumulation of $\beta$-carotene in the stratum corneum. This is likely secondary to reduced hepatic conversion of $\beta$-carotene to vitamin A in the setting of hypothyroidism, leading to increased circulating carotene levels.

## Myxedema

**Myxedema** is the hallmark cutaneous finding in hypothyroidism and is distinct from the dermopathy associated with Graves disease. It results from dermal deposition of **mucopolysaccharides**, particularly **hyaluronic acid** and **chondroitin sulfate**. This accumulation leads to skin that is **doughy, swollen, and waxy**, without pitting edema. Myxedema is typically generalized but may be more pronounced in the extremities.

With treatment of the underlying hypothyroidism, myxedema tends to resolve.

### *Characteristic Facial Features*

- **Broadened nose**
- **Thickened lips**
- **Puffy eyelids**

- **Macroglossia** (enlarged, smooth, clumsy tongue)
- **Loss of facial expression** (a pathognomonic feature)

### *Cutaneous Findings in Hypothyroidism*

- Doughy, dry skin with fine wrinkling
- Yellow-orange carotenemia
- Dry, brittle hair with slow growth
- Brittle nails with longitudinal and transverse striations
- Impaired wound healing

### *Hair Changes*

- Hair becomes **coarse, dry, and brittle**
- **Diffuse or patchy scalp hair loss**
- Increased proportion of hair follicles in the **telogen phase**
- **Telogen effluvium** may occur following acute onset
- **Loss of the outer third of the eyebrows** (see Fig. 152-12)
- While body hair may be reduced, there can be paradoxical increase in **lanugo-type hair** on the back, shoulders, and extremities

### *Nail and Oral Findings*

- Nails grow slowly and may become **thickened, brittle**, and show **longitudinal or transverse ridges**
- **Macroglossia** contributes to a large, smooth, red, and clumsy tongue
- Hoarseness may occur due to **recurrent laryngeal nerve compression** in patients with goiter

### *Goiter-Related Complications*

- Large goiters can cause **tracheal or esophageal displacement**

- **Retrosternal goiters** may lead to **Pemberton's sign**: facial plethora and jugular venous distention when arms are raised above the head, indicating thoracic inlet obstruction

## Diagnosis

- **Serum thyrotropin (TSH)** is the best initial test
- Additional tests: **free triiodothyronine (T3)** and **free thyroxine (T4)**

## Management

- **Endocrine consultation** recommended
- **Oral thyroxine replacement** (typically 75–150 µg/day)
- **Careful monitoring** for coexisting heart, lung, or adrenal disease

▲ FIGURE 152-12 Hypothyroidism. This patient exhibits classic signs including dry, pale skin; loss of the lateral third of the eyebrows; facial puffiness due to dermal mucopolysaccharide accumulation; broadened nose; macroglossia; drooping eyelids; and diminished facial expression—the most pathognomonic feature.